

Exponent Simplification Exercises

Rules for this exercise:

- Only integer exponents, no fractional exponents and no radicals.
- Negative exponents may appear during calculation, but final answers must contain no negative exponents.
- if ans like this x^y , you just write x^y
- if ans like this $\frac{x}{y}$, you just write $\frac{x}{y}$

1 1 Basic Problems

Simplify each algebraic expression:

1. $x^5 \cdot x^3$
2. $a^{10}a^4$
3. $(m^4)^3$
4. $(4y)^2$
5. n^0 ($n \neq 0$)
6. $\frac{k^3}{k^8}$
7. $(3ab)^3$
8. $\frac{p^2}{p^9}$

2 2 Intermediate Mixed Problems

1. $x^4 \cdot xx^7$
2. $(2c^3)^2 4c^4$
3. $\frac{(5t)^2}{t^6}$
4. $\left(\frac{x^4}{y^2}\right)^3 x^{15}$
5. $a^2 \cdot a^5 a^{10}$

$$6. \frac{(2s^2)^4}{16s^9}$$

$$7. (3m^2n)^2 m^7$$

$$8. \frac{z^4 \cdot z}{z^{11}}$$

3 3 Advanced Comprehensive Problems

$$1. \frac{(2a^2b)^3}{(4a^4b^2)^2}$$

$$2. \left(\frac{3x^2}{y^3}\right)^2 x^9$$

$$3. \frac{5^{n+4}}{5^{n+7}}$$

$$4. (2p^3q)^2 \cdot pp^{10}$$

$$5. \frac{(3m^4)^2}{9m^{11}}$$

4 4 Evaluation Problems

Given $a^m = 5$, $a^n = 3$

$$1. \text{ Evaluate } a^{m+n}$$

$$2. \text{ Evaluate } a^{3m-n}$$